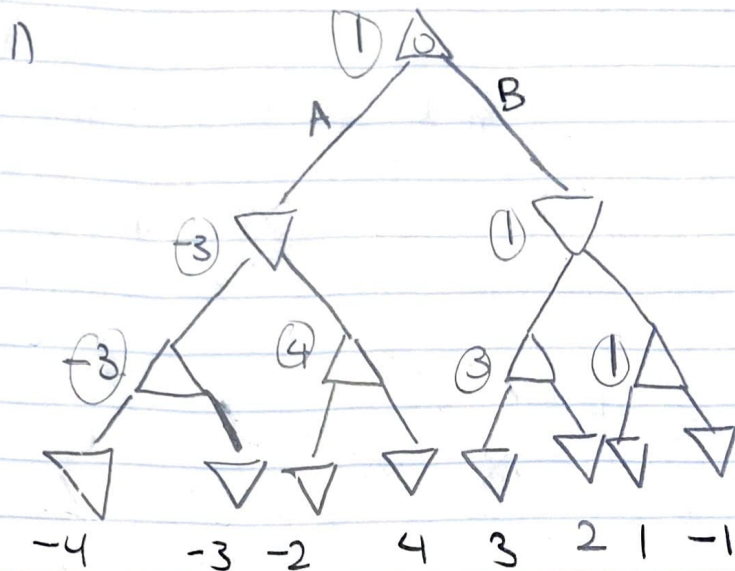
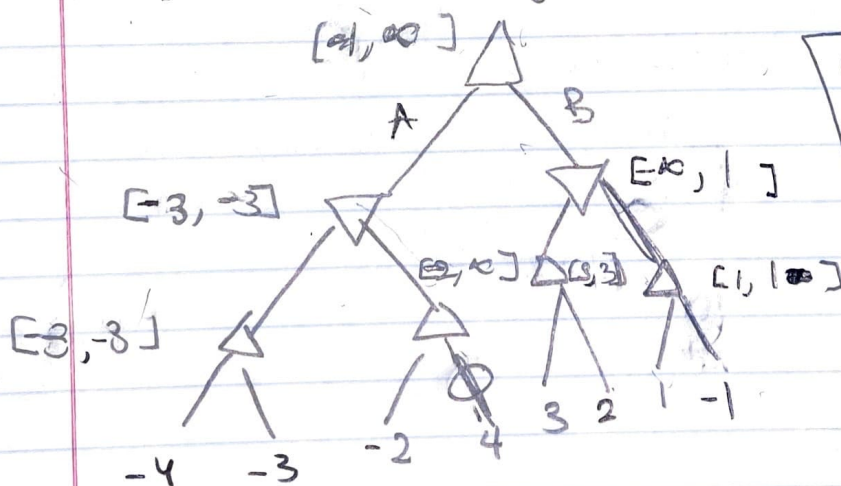




Player I should take action B as it is the most optimal choice for him, as a max player, because it leads to highest minimax value.

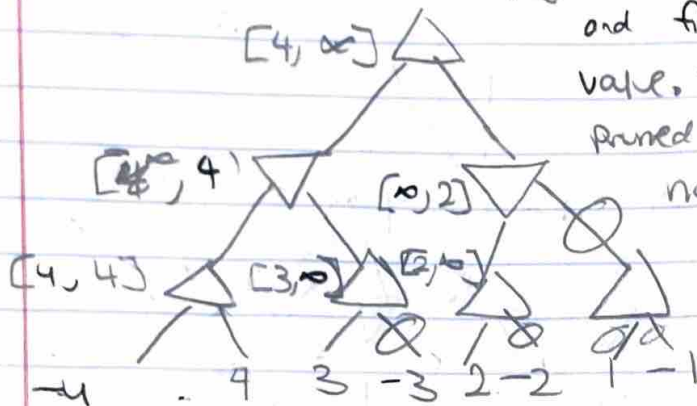
The expected outcome is getting the value 1.

Using alpha beta pruning, we get a tree like this

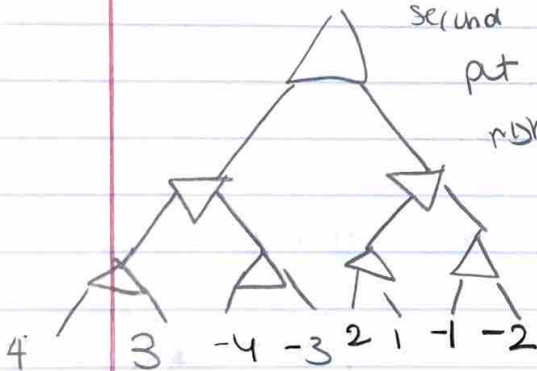


Only branch for leaf of value 4 is pruned because $-3 < -2$

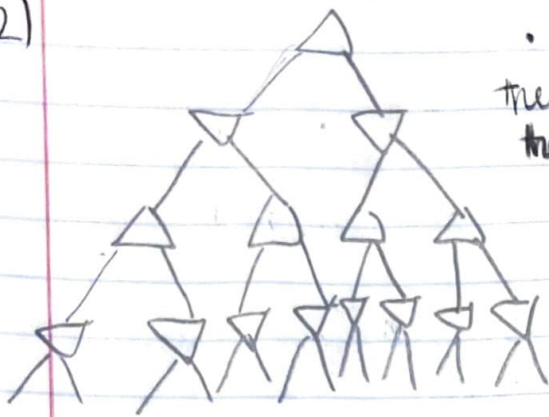
To maximize number of nodes pruned, I would reorder them so each max node wins in third level and first left one has largest value. Therefore, entire subtree pruned as well as two more nodes.



To minimize node pruning, we have to make the pruning algorithm check every node. ~~Since~~ Since second level is a min layer, we put all the negative numbers in the right side.



2)



. I don't think the player at the root can force a win because the other player has last choice so ~~he~~ he can always decide between a losing state or a tie before a win.

I think ~~it~~ it doesn't matter as if the 2 non zero states are adjacent, the player can have a direct choice between them but if they are far apart one of those states can be avoided from choosing an opposite side at the root.

I think if ~~the~~ the depth of the tree was odd instead of even, it will change. This is because the root player plays first and last and can force himself a win. ~~He can~~ It still matters if the non-zero states are located but since the root player has a better control, it matters slightly less.

3a

vars = { A, B, C } domain for

domains: $\text{dom}(\text{var}) = \{0, 1, 2\}$ (index of their order)

constraints

$\text{dom}(A) \neq \text{dom}(B)$

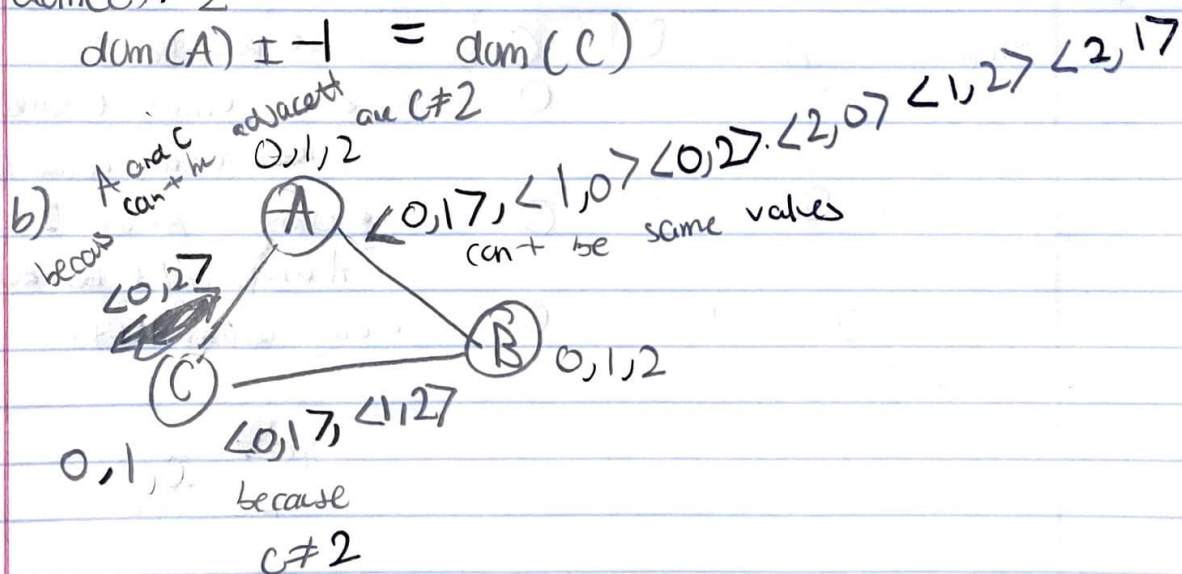
A and C are PhD's, can't also have same index

B is a MS

adjacent can't have same degree $\Rightarrow$

$\text{dom}(C) \neq 2$

$\text{dom}(A) \pm 1 = \text{dom}(C)$



c)

step	A	B	C	explanation
0	0	1		choose 1 $\Rightarrow A \neq B$
0	0	1	2	backtrack, $C \neq 2, B = 2$
0	0	2		
0	0	2	1	backtrack A and C are adj
1	1	0		
1	1	0	2	backtrack, $C \neq 2$
1	1	2		
1	1	2	0	back track A and C are adj
2	2			
2	2	0	1	backtrack, A and C are adj
2	2	0	1	

step	A	B	C	explanation
	2			
	2	1		
	2	1	0	works!

Backtracking happens when A and C were adjacent to each other or when C was put last in order

D)

step	A	B	C	explanation
			0	• we choose C since it has less remain because $C \neq 2$
	2		0	• chose A as 2 because A and C can't be adj.
	2	1	0	• only option left.